Diversification-Based Portfolio Construction Using CRSP Data

COMP 462 — Optimization Models in Finance

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1 Introduction and Project Goal

In this project we tested the portfolio optimization methods shown in class on a portfolio of stocks that we sampled from the CRSP database based on monthly returns and market capitalization data. We built entirely invested and long-only portfolios using:

1. Minimum Variance (MinVar),
2. Maximum Diversification (MaxDiv),
3. Risk Parity (RiskParity).

We compared their performance against:

1. Value-Weighted (ValueWeighted),
2. Equal-Weighted (EqualWeighted).

We evaluated the strategies using a rolling backtest from January 2017 to December 2022.

2 Data Description

We used three data sources:

- **CRSP**: monthly total returns, market capitalizations, PERMCO identifiers, and tickers for individual stocks.
- **Market Return Data**: monthly total market returns.
- **Risk-Free Rate Data**: monthly 30-day Treasury bill returns.

We computed the monthly excess returns by

$$r_{i,t}^{(e)} = r_{i,t} - r_{f,t}, \qquad r_{m,t}^{(e)} = r_{m,t} - r_{f,t},$$

where $r_{i,t}$ is the stock return, $r_{m,t}$ is the market return, and $r_{f,t}$ is the risk-free rate.

3 Backtest Design

We used a rolling-window backtesting procedure. It works like this:

- Start date: January 2017
- $T = 60$ months (estimation window)
- Universe size: $N = 30$ stocks
- Rebalancing frequency: Monthly
- Out-of-sample period: January 2017 – December 2022

At each out-of-sample month t:

1. We estimated the model parameters using the past 60 months.
2. We computed the portfolio weights for each of the strategies.
3. We recorded the realized excess return in month $t + 1$.

4 Universe Selection

We excluded any stock that did not have a full 60-month history of returns prior to January 2017. Among the remaining stocks we selected the top 30 by market capitalization at that date. This fixed 30-stock universe is held constant during the evaluation period.

5 Covariance Estimation

To estimate the covariance matrix without requiring massive overhead and compute time, we used a single-factor market model. For each stock i we fit

$$r_{i,t}^{(e)} = \alpha_i + \beta_i r_{m,t}^{(e)} + \varepsilon_{i,t}.$$

Using OLS estimation, we obtained β_i and the residual variance $\sigma_{\varepsilon,i}^2$. Let σ_f^2 be the variance of the market factor. The covariance matrix is

$$V = \sigma_f^2 \beta \beta^\top + D,$$

where D is a diagonal matrix containing the idiosyncratic variances.

6 Portfolio Construction

All portfolios are long-only and fully invested:

$$\mathbf{1}^\top x = 1, \quad x \geq 0.$$

6.1 Minimum Variance Portfolio

The minimum variance portfolio solves

$$\min_x \; x^\top V x \quad \text{s.t. } \mathbf{1}^\top x = 1, \; x \geq 0.$$

6.2 Maximum Diversification Portfolio

The diversification ratio is

$$DR(x) = \frac{x^\top \sigma}{\sqrt{x^\top V x}},$$

where σ is the vector of individual asset volatilities. The MaxDiv portfolio maximizes this ratio subject to the same long-only, fully-invested constraints.

6.3 Risk Parity Portfolio

Risk parity equalizes contributions to portfolio variance:

$$RC_i(x) = x_i (Vx)_i, \quad RC_1 = RC_2 = \cdots = RC_N.$$

We used an iterative proportional update procedure to compute the long-only risk parity weights.

6.4 Benchmark Portfolios

- **Equal-Weighted**: $x_i = 1/N$.
- **Value-Weighted**: x_i proportional to market capitalization.

7 Performance Metrics

Performance is evaluated using monthly excess returns. Reported metrics include:

- Annualized return: $12 \times \bar{r}$.
- Annualized volatility: $\sqrt{12} \times \sigma(r)$.
- Sharpe ratio: AnnRet/AnnVol.
- Maximum drawdown: largest peak-to-trough decline in the cumulative wealth index.

8 Results

8.1 Summary Statistics

Table 1 reports annualized out-of-sample performance statistics.

Table 1: Out-of-Sample Performance Summary (2017–2022)

Strategy	Return	Volatility	Sharpe	Max Drawdown
MinVar	12.4%	11.9%	1.05	-10.5%
MaxDiv	15.8%	11.9%	**1.33**	-12.2%
RiskParity	14.4%	13.8%	1.04	-14.8%
ValueWeighted	**19.3%**	17.4%	1.11	-19.9%
EqualWeighted	14.9%	15.7%	0.95	-17.6%

8.2 Risk–Return Tradeoff

To visualize the risk–return tradeoff, Figure 1 plots annualized excess returns against realized volatility for each strategy. For each point we display both the average excess return (solid dot) and the compounded excess return over the sample ("×" marker), which is slightly lower because of volatility drag.

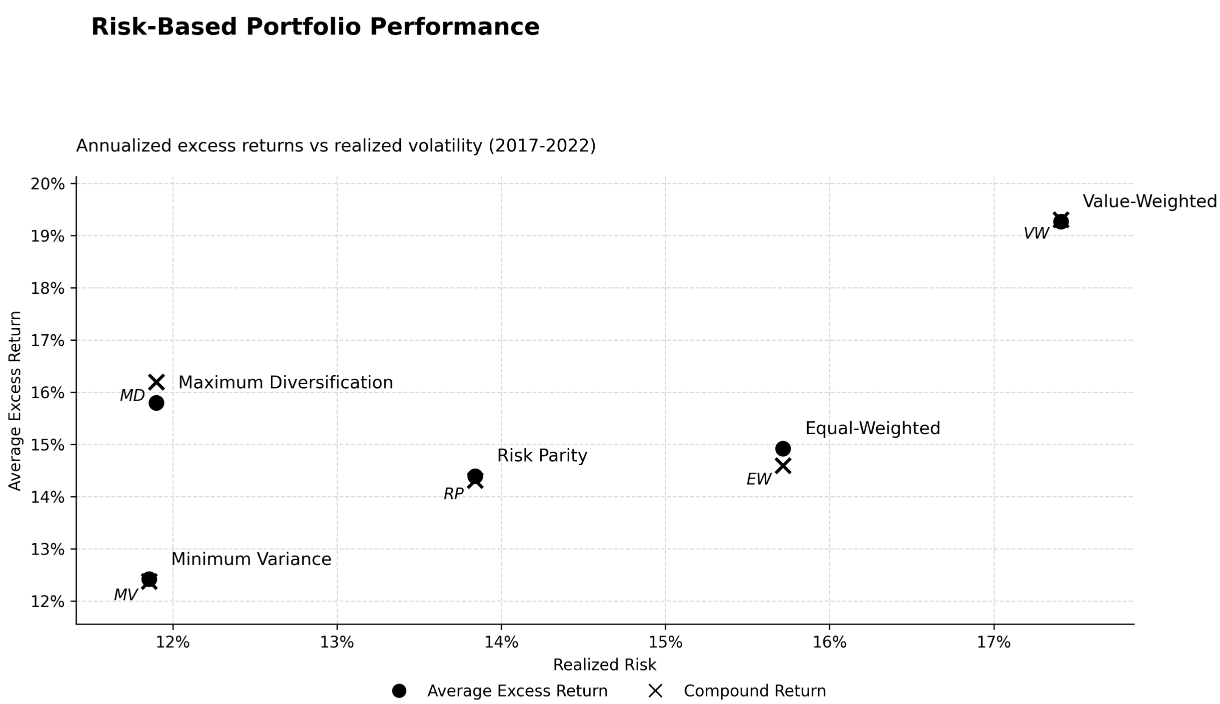

Figure 1: Risk-Based Portfolio Performance: Annualized Excess Return vs. Realized Volatility (2017–2022)

The scatter plot highlights several features that are less obvious from the table alone. MaxDiv lies furthest up and to the left among the optimized strategies: it achieves a higher excess return than MinVar and RiskParity at essentially the same or lower volatility, which is consistent with its leading Sharpe ratio. ValueWeighted sits at the far right with both the highest risk and the highest return. This is because of it's high percentage in high market cap stocks, but since it is so concentrated in those stocks, it introduces a lot more risk as well. EqualWeighted lies between ValueWeighted and the risk-based portfolios in both dimensions, having a pretty high risk for mediocre return. MinVar lays in the low-risk, low-return corner of the figure, and Risk Parity has a little bit more risk and return than minimum variance, but with less excess return than something like MaxDiv. Taken together, Figure 1 suggests that MaxDiv is the most efficient of the optimized strategies in terms of getting the best return for the amount of risk you are putting in, whereas ValueWeighted dominates in raw return percentage but at the cost of substantially higher volatility.

8.3 Cumulative Wealth

Figure 2 shows cumulative wealth curves starting at 1 in January 2017.

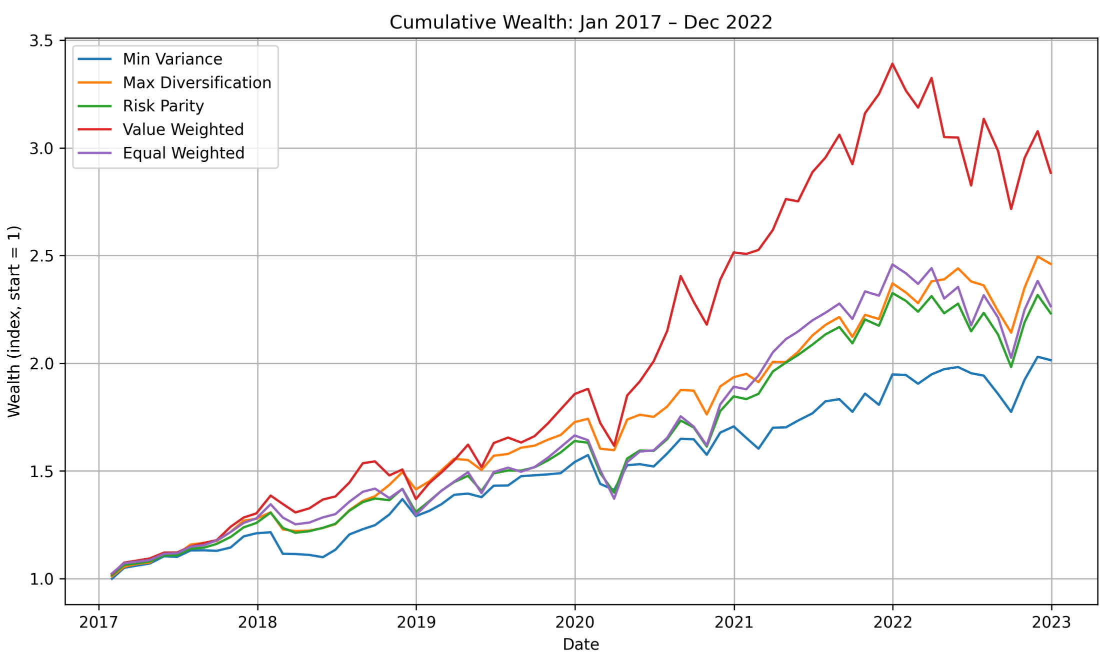

Figure 2: Cumulative Wealth: January 2017 – December 2022

The wealth paths confirm the patterns in Table 1 and Figure 1. ValueWeighted compounds to the highest terminal wealth, driven largely by the strong performance of mega-cap growth stocks during 2017–2021, but also experiences the deepest drawdowns in 2020 and 2022. MaxDiv and EqualWeighted track each other closely for much of the period, with MaxDiv generally on top, while MinVar produces the smoothest but lowest trajectory. RiskParity sits between the defensive MinVar path and the more aggressive VW/EW paths.

8.4 Portfolio Weights

Figure 3 shows the portfolio weights for MinVar, MaxDiv, and RiskParity at the final rebalancing date, using stock tickers for interpretability.

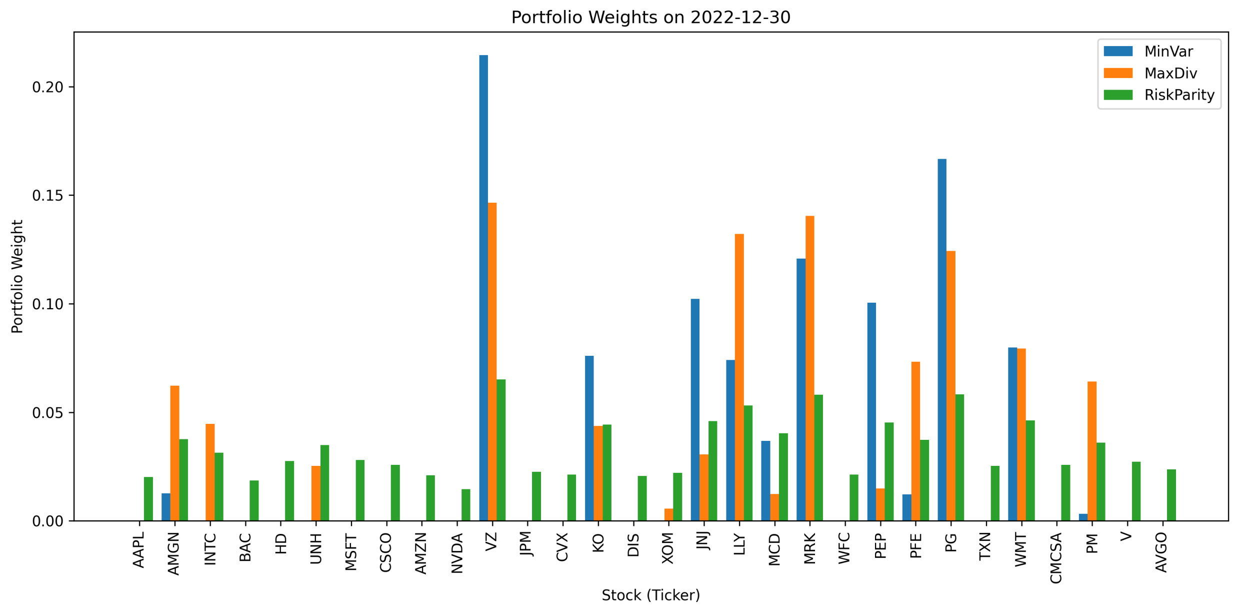

Figure 3: Portfolio Weights on December 30, 2022

The cross-sectional allocations reveal important qualitative differences across the diversification rules. The MinVar portfolio is extremely concentrated in a handful of defensive, low-beta names. It assigns very large weights to Verizon (`VZ`), Procter & Gamble (`PG`), Johnson & Johnson (`JNJ`), Merck (`MRK`), Walmart (`WMT`), and PepsiCo (`PEP`), all of which are classic low-volatility, stable-cash-flow companies in telecom, consumer staples, and healthcare. At the same time, MinVar puts almost no weight on high-beta growth stocks such as Apple (`AAPL`), Amazon (`AMZN`), Nvidia (`NVDA`), Broadcom (`AVGO`), and Visa (`V`), despite their large market capitalizations in the ValueWeighted benchmark.

The MaxDiv portfolio shares a similar safe tilt, but is less extreme. It still overweights `VZ`, `PG`, `JNJ`, and `MRK`, reflecting the attractive diversification properties under the single-factor risk model, but it spreads capital across a broader set of names, including Amgen (`AMGN`), Intel (`INTC`), Coca-Cola (`KO`), Eli Lilly (`LLY`), and McDonald's (`MCD`). This produces a more balanced allocation that avoids the single-name concentration observed in MinVar, while also getting any potential upside of one of those broader set of names going up significantly.

RiskParity is the most evenly distributed of the three. Nearly every stock receives a weight between roughly 3% and 6%. RiskParity still mildly tilts toward the same defensive names. For example `PG`, `JNJ`, `MRK`, and `WMT` receive weights above the equal-weight level of $1/30 \approx 3.3\%$, while underweighting more volatile names like `AAPL`, `AMZN`, `NVDA`, and `AVGO`. Because of this no single ticker dominates the portfolio, which aligns with the goal of equalizing risk contributions rather than capital weights.

These differences in holdings help explain the performance patterns. ValueWeighted relies heavily on the large-cap technology names that led the market during this period, boosting its compounded return, but also exposing it to sharper drawdowns when those few high-performing names sold off. By design MinVar, sacrifices exposure to the mega-cap winners in order to minimize variance, resulting in a smoother but lower-growth path. MaxDiv and RiskParity sit between these extremes. They both retain some exposure to growth stocks, but still by design tilt toward low-volatility, low-correlation sectors such as consumer staples and healthcare.

9 Discussion

The combination of the Table 1, Figure 1, Figure 2 and Figure 3 creates a consistent image of the way the various diversification strategies act in this universe.

The first is that MaxDiv generates the best Sharpe ratio and is pretty good in terms of the risk-return plane. It is above MinVar and RiskParity at virtually the same or lower volatility. This implicates that the explicit maximization of the diversification ratio is an effective exploitation of the low-correlation covariance matrix structure, especially between defensive stocks (PG, JNJ, and MRK). MaxDiv also has a wider range of holdings than MinVar, not subjecting itself to excessive single-name investments and at the same time focusing on areas that shield the market factor.

Second, the ValueWeighted benchmark is the best in terms of raw return at the expense of significantly more volatility and drawdowns. The 2017-2021 market regime explains its performance, where mega-cap grows and technology stocks, such as AAPL, MSFT, AMZN, NVDA, and AVGO, greatly outperformed the market at large. Our optimized portfolios, and especially MinVar and MaxDiv, underweight these names, in relation to lower-beta defensives, that increases their risk-adjusted performance and do not allow them to match VW in performance in terminal wealth.

Third, MinVar portfolio acts as a defensive strategy. It has the lowest maximum drawdown and volatility in the sample, and cumulative wealth path that is smoother than the others. MinVar's heavy concentration in consumer staples and healthcare names increases robustness to market selloffs; however, the lack of exposure to the leading growth stocks results in lower average and compounded excess returns.

Lastly, there is RiskParity which offers an intermediate. It generates weights by leveling the risk contributions and produces a nearly uniform cross-sectional weight profile only with weak sector tilts. In the risk-return plot it lies between MinVar and EqualWeighted, and in cumulative wealth is the middle course. A balanced portfolio would be expected to provide trade-offs between aggressive cap-weighting and extremely defensive minimum-variance positioning.

On the whole, our findings emphasize the fact that diversification strategies relative performance depends on the existing market regime and to the exposure each approach performs at the stock level.

10 Limitations and Extensions

This research has a number of limitations. To begin with, we use a single-factor covariance model; style and sector factors that include size, value, momentum, and industry effects are excluded. A multi-factor risk model may alter the estimated correlations, and thus optimize the weights. Second, our portfolio is limited to 30 big U.S. stocks which limits the possibility of diversification and can overstate the influence of such personal names as AAPL or PG. Third, we disregard transaction costs and trading frictions. EqualWeighted and the optimized portfolios are some of the strategies that need regular rebalancing, net-of-cost performance would be lower, especially of higher-turnover allocations. Lastly, our findings are based on the selection of a 60-month estimation window and monthly rebalancing. The possible problem with this is that other windows or frequencies may result in dissimilar risk values and, as a result, different levels of performance.

Possible extensions could include implementing a multi-factor risk model, explicitly modeling transaction costs, and repeating the analysis on larger universes and different historical periods to assess robustness across market regimes.

11 Conclusion

Using the results of our empirical experiment, it is evident that optimization based diversification schemes can be better than trivial weighting methods, although what "better" means depends on the objective of the investor.

Maximum Diversification has the best performance has the best risk-adjusted performance, as it has low-risk without having the extreme low-cap concentrations of MinVar. This allows for upside from riskier stocks, which is why maximum diversification had the best sharpe ratio out of the ones we tested.

Minimum Variance is best on downside coverage and the has the smoothest upward return by focusing on low-volatility, consistent names. But as a result it sacrifices the high growth that mega cap stocks like the Mag7 could potentially have in bull markets (like from 2017-2022) that models like ValueWeighted capitalized on heavily. Risk parity has an equal or balanced allocation which offers moderate risk and also moderate returns.

All these findings emphasize the significance of the covariance structure and stock-level exposures in the decision-making of portfolios. Specifically, the way each diversification method allocates money between low-beta sectors (like telecom, healthcare, etc.) and high-beta growth-oriented technology technology and consumer-discretionary names, is critical for understanding why the strategies performed the way they did over 2017-2022 (and why they could perform differently under different economic conditions).